W-W-650C
June 22, 1970
SUPERSEDING
Int. Fed. Spec. W-W-00650b(NAVY-Ships)
October 26, 1966 and
Fed. Spec. W-W-650a
July 29, 1957

FEDERAL SPECIFICATION

WRENCH, IMPACT, ELECTRIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable electric impact wrenches designed to impart a rotary movement of force to wrench sockets or screw driver bits.

1.1.1 Federal specification coverage. This specification does not include all sizes and capacities of electric impact wrenches, but is intended to cover only those sizes and capacities generally used by the Federal Government.

1.2 Classification.

1.2.1 Sizes and capacities. Impact wrenches shall be of the following sizes and nominal capacities, as specified (see 6.2):

Size 1 - 3/8 inch square drive - 1/4-inch bolt capacity.
Size 2 - 1/2 inch square drive - 3/8-inch bolt capacity.
Size 3 - 5/8 inch square drive - 5/8-inch bolt capacity.
Size 4 - 3/4 inch square drive - 7/8-inch bolt capacity.
Size 5 - 1 inch square drive - 1-1/4-inch bolt capacity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distriubtion points in their agencies.)

Military Specifications:

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For the Treatment of Communications Electronic and Associated Electrical Equipment).

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Military Standards:

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-1337 - Electric Portable Hand Tools General Suppression Systems Design Requirements.

(Copies of Military Specifications and Standards required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids, or request for proposal shall apply:

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standards for Federal Services.

(Application for copies shall be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

American National Standards Institute (ANSI), Inc. Standards:

B3.5 - Tolerances for Ball and Roller Bearings.
B3.6 - Boundary Dimensions for Ball and Roller Bearings.
B3.8 - Mounting Ball and Roller Bearings, Specifications for.
B5.38 - Driving and Spindle Ends for Portable Air and Electric Tools.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

Underwriters' Laboratories (UL), Inc. Standards For Safety:

No. 45 - Electric Tools.
No. 498 - Attachment Plugs and Receptacles.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, L. I., N.Y. 11749; 1655 Scott Blvd., Santa Clara, California 95950.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for the convenience of identification, and are not intended to preclude the purchase of impact wrenches which are otherwise in accordance with the requirements of this specification.

3.2 First article inspection. When specified (see 6.2), the contractor shall furnish a completely fabricated electric impact wrench for first article inspection in accordance with section 4. The sample shall be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used in production of items to fulfill the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirement of this specification.

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3.3 Material. All materials used in the fabrication of the impact wrenches shall be new and of a quality necessary to produce an impact wrench which will meet the requirements described herein.

3.4 Design. The impact wrenches and accessories furnished under this specification shall be new, of the manufacturer's latest design and shall provide convenient and safe operation. All parts subject to wear, breakage or distortion shall be easily accessible for adjustment or replacement.

3.4.1 Safety. Electric impact wrenches shall be designed for facility in operation and safety in use. All rotating parts which are hazardous to the operator shall be suitably guarded against accidential contact. Guards shall not interfere with the satisfactory operation of the tool. The housing, except for ventilating openings, shall totally enclose all stationary and rotating electrical parts. Wrenches with suppression systems (see 6.2), shall bear a permanent label warning of the necessity for using the grounding connection to avoid shock. In addition to protection from moving parts, the wrench shall be designed to prevent electrical shock to the operator. The wrench shall be so constructed and ventilated that the external surfaces do not attain uncomfortable temperatures during test to determine temperature rise in compliance with UL45 as specified in section 4.

3.4.2 Lubrication. All bearings (except sealed-for-life type), and sliding parts shall be provided with means for lubrication. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.4.3 Interchangeability. All parts shall be manufactured to standards which will permit replacement or adjustment without modification. All parts having the same manufacturer's part number shall be completely interchangeable with each other with respect to size, shape and performance.

3.4.4 Conformance to standards of Underwriters, Inc. The contractor shall submit to the contracting agency proof that the basic wrench he proposes to supply under this specification conforms to the requirements of UL45. The label, or listing with reexamination of the UL may be accepted as evidence that the wrench conforms to this requirement. In lieu of the label, or listing with reexamination, the contractor may submit independent proof, satisfactory to the contracting agency, that the wrench conforms to the applicable requirements of the published standards including methods of tests of the UL noted above. Compliance with requirements of UL regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his wrench. In event of difference between any of the requirements of this specification and those of UL, the requirements of this specification shall govern.

3.5 Construction. The impact wrench shall be free of any characteristics or defects which will prevent passing the examinations or tests specified in section 4. The impact wrench shall be complete so that when connected to the specified source of power, it can be used for any operation for which it was designed.

3.5.1 Screws, bolts and nuts. Steel through bolts with nut shall be used wherever practicable for securing aluminum alloy parts to avoid threading the aluminum alloy. Screws or studs, if used, shall engage the threads of the parts secured for a distance equal to at least twice the outside diameter of the screw or stud, or be equipped with ferrous inserts. Lock washers shall be provided for all screws, studs, and bolts. Threaded parts shall be in the inch system and conform to H28.

3.6 Surface finish. Exposed surfaces shall be smooth, edges shall be rounded or beveled. Surfaces shall be finished according to the manufacturers standard practice.

3.7 Sizes and capacities. Electric impact wrenches furnished under this specification shall be similar to figure 1 for sizes 1, 2, 3, and 4; figure 2 for size 5, and shall conform to the capacities, dimensions, and weight specified in table I for the size specified (see 6.2).

3.8 Components. Impact wrenches furnished under this specification shall consist essentially of wrench housings, impacting mechanism, spindle, motor and electrical components and accesscries.

3.8.1 Housings. Wrench housing shall be high strength cast aluminum alloy, steel or equally suitable materials having characteristics as to withstand the stresses imposed by normal impacting and during continuous use. Housings shall include the motor case, impacting mechanism, spindle housing and handles.

3.8.2 Impacting mechanism. Impact wrenches shall include an impacting mechanism designed to withstand long continuous use and capable of delivering a rotary impact movement of force to the spindle. Impacting mechanism parts such as the hammer, the anvil and related components shall be alloy steel suitably hardened for heavy-duty continuous service. Impacts per seconds and ultimate torque after 5 minutes of impacting shall conform to table I for the size wrench specified.

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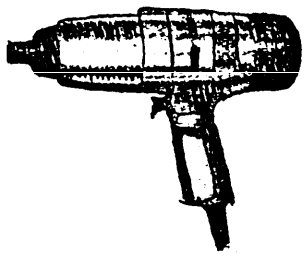


FIGURE 1. Wrench impact electric sizes 1, 2, 3,and 4.

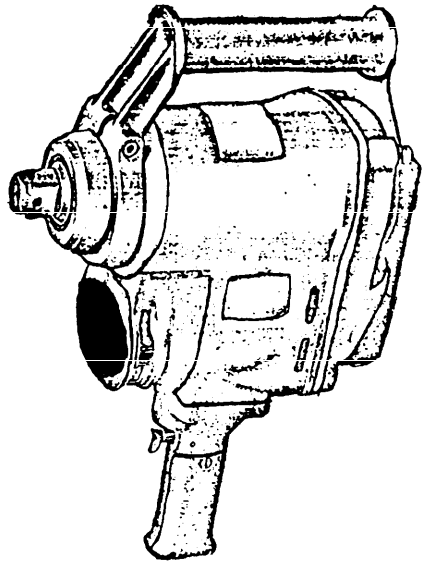


FIGURE 2. Wrench impact electric size 5.

TABLE I. Impact wrench requirements

Requirements	Size 1	Size 2	Size 3	Size 4	Size 5
Size of square drive	3/8 inch	1/2 inch	5/8 inch	3/4 inch	1 inch
Rated bolt size capacity inches (min.)	1/4 inch	3/8 inch	5/8 inch	7/8 inch	1-1/4 inch
Ultimate torque after 5 seconds impacting (foot, pounds (min.)	40	100	200	300	850
Free speed RPM (min.)	1700	1750	1600	1600	800
Impacts per minute (min.)	2000	1700	1600	1600	800
Weigh less cable lbs. (max.)	5	8	12	17	35
Spindle offset inches (max.)	1-1/2	1-3/4	2	2	2-1/2

3.8.3 Spindle end. Wrenches shall be furnished with a square drive spindle end. The square drive shall be hardened to 57 to 60 Rockwell C and shall conform to B5.38. When specified (see 6.2), wrenches shall be furnished with a 7/16-inch hexagonal female chuck conforming to B5.38 and a square drive adapter having a 7/16-inch male hexagon shank to fit the chuck, and a square drive of the size specified (see 6.2).

3.8.3.1 Friction retainer type. As an alternate method, the square end of the spindle may utilize a friction spring type retainer. When utilized, the dimensions of the square drive shall be such as to adequately retain female square drive sockets conforming to B5.38.

3.8.4 Motor. Motor shall be universal type designed to operate on either 115 volt, direct current or 115 volt (± 6 percent) single phase, 25-60 cycle alternating current. Motor shall be mounted integral with the tool. Access to the brush rigging, brushes and commutator shall be provided. Motor shall comply with the requirements of UL45, for industrially rated tools. Rotors and fans shall be accurately balanced dynamically within standard limits specified commercially for the motor size used.

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3.8.4.1 Commutators. Commutators shall have "V" ring and sheet mica or molded construction. Molded commutators shall be of material that will withstand maximum rotational and impacting forces and be immune to deformation from temperatures encountered in service.

3.8.4.2 Reversing mechanism. Motors shall be equipped with a suitable brush shifting mechanism or a double pole, double throw switch for reversing the direction of rotation. The reversing mechanism shall be easily operable by the operator.

3.8.4.3 Electrical equipment. All electrical components including motor, switch, wiring, flexible cord (cable) attachment plug cap and equipment grounding shall conform to applicable requirements of UL45. The cord shall have an attachment plug cap and unless otherwise specified (see 6.2), a minimum free length of 9 feet measured from point of entry into the impact wrench. Attachment plug cap shall conform to UL498 requirements for 2-pole, 3-wire grounding devices rated 15 amperes at 125 volts.

3.8.4.4 Electromagnetic interference suppression. When specified (see 6.2), each impact wrench shall be electromagnetic interference suppressed in accordance with the requirements of MIL-STD-1337.

3.8.4.5 Fungus proofing. When specified (see 6.2), the electrical circuitry, including all components and connections, shall be protected from the effect of water and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, with the following exceptions:

(a) Components or circuit elements which are inherently fungus and water resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied to give a minimum dry film thickness of 0.001 inch to component or element surface previously cleaned and prepared in accordance with accepted commercial practice.

3.8.4.6 Bearings. Armature shaft bearings shall be of the ball or roller type, and shall be of the dustproof, sealed construction or shall be mounted in a dustproof housing. Intermediate shaft bearings shall be either ball, roller, needle, or sleeve type. Steel inserts may be omitted in the mounting of bearings when sufficient contact area is provided on the mounting surface. Roller and needle bearings which are designed primarily for radial loads shall not be used to absorb thrust loads. Bearings where required to absorb axial thrust loads must be provided for by use of a separate thrust bearing, either ball, roller or an intergral combination type roller thrust bearing. Ball and roller bearings shall conform to B3.5 for tolerances, B3.6 for boundary dimensions and B3.8 for mounting requirements.

3.8.4.6.1 Spindle bearings. The spindle bearings shall be of the externally sealed roller or sleeve type. The bearing shall be capable of withstanding side thrust and impacting forces.

3.9 Performance. The wrenches shall transmit no objectionable kick or twist to the operator under any operating conditions. The wrenches shall be capable of withstanding continuous service under operating conditions to which wrenches of this type are normally subjected, and comply with the applicable test for impact wrenches as specified in UL45. Wrenches shall conform to table I for the wrench size and capacity specified, when tested as specified in section 4 for performance.

3.10 Carrying case. When specified (see 6.2), a durable carrying case of sufficient size to contain the impact wrench, maintenance tools and accessories shall be furnished. The case shall be complete with hinged cover, carrying handles and closing fastener. The interior and exterior surfaces of the carrying case shall have a durable coating of paint or enamel, except surfaces which are inherently corrosion free or electroplated. Case shall be marked or lettered to show the following:

(a) "Wrench, Impact, Electric, Portable".
(b) Stock Number FSN 5130-...-...
(c) Drive square size -
(d) Bolt capacity-
(e) Electrical characteristics-

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3.11 Identification plate. Each wrench shall be permanently marked by die stamping, casting or furnished with an identification plate to include the following:

(a) Manufacturer's name or trademark.
(b) Manufacturer's model or serial number.
(c) Electrical characteristics (volts, phase, cycles).
(d) Rated load - Amperes.

3.12 Handbook of instructions and parts list. Unless otherwise specified (see 6.2), the manufacturer shall furnish one (1) copy of his standard instruction book and parts list with each impact wrench. The book shall include illustrations, instructions and data necessary for safe operation, maintenance adjustment and repair. The parts list shall include the manufacturers part number and ordering instructions.

3.13 Accessories. Each impact wrench shall be furnished with such accessories, attachments and tools as normally furnished commercially. Additional accessories shall be furnished, as specified (see 6.2).

3.14 Spare parts. Additional spare parts shall be furnished, as specified (see 6.2). When specified (see 6.2), each wrench shall be furnished with the following spare parts:

1 each - Anvil.
1 each - Trigger switch.
1 each - Reversing mechanism contacts or switch.
2 each - Socket retainers.

3.15 Workmanship. Workmanship of the impact wrenches shall be of a quality prevailing among manufacturers normally producing impact wrenches of the sizes and capacities specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection.
(b) Quality conformance inspection.

4.2.1 First article inspection. First article inspection shall be applied to the first article submitted in accordance with 3.2. Failure of the first article to pass the examination of one or more preproduction tests shall be cause for rejection. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.4.1 and all tests in 4.5.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All impact wrenches of the same size shall be considered a lot for purposes of quality conformance inspection.

4.3.2 Sample. A random sample shall be selected from each lot in conformance with MIL-STD-105 at inspection level II.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item selected in accordance with 4.3.2 prior to being offered for acceptance under the contract. Quality conformance inspection shall consist of the following:

(a) Examination (4.4.1).
(b) Tests (4.5.1 thru 4.5.3).
(c) Inspection of preservation, packaging, packing and marking for shipment and storage (4.6).

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4.4.1 Examination. Each sample selected in accordance with 4.3.2 shall be examined for design, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with table II. The acceptable quality level (AQL) for major defects shall be 2.5 percent defective and for minor defects 6.5 percent shall not be included in lot for delivery and if the number of rejects exceeds the acceptance number for that sample, the lot represented shall be rejected.

TABLE II. Classification of defects

Examine	Defects	AQL percent defective	
		Major 2.5	Minor 6.5
Design	Capacity or size of wrench less than specified.	X	
Material	Not in accord, when specified.	X	
Construction	Wrench incomplete, parts missing or not securely assembled.	X	
	Component parts defective; bent, cracked, split or not free from sharp edges, fins, or burrs.	X	
	Motor not type specified (see nameplate).	X	
	Attachment plug not type specified, defective, not secure on cord, prongs, loose or broken.	X	
	Switch defective, loose, sticks, or binds not rigidly mounted, switch not specified type, or switch (as applicable).	X	
	Exceeds weight specified in table I.	X	
	Cord defective, jacket cracked, split, or bruised: Cord length not as required, cord not provided with an anti-kink and strain relief device at junction with wrench, type of cord not as specified.	X	
	Exceeds spindle offset specified in table I.	X	
	Not provided with reversing mechanism or mechanism not operable.	X	
Technical manuals	Instruction book and parts list missing, illegible, or does not include information required.		X
Marking	Indentification plate not securely fastened to frame or housing. Plate stamping or cast imprints missing, illegible or incorrect.		X

4.4.2 Tests for quality conformance. Two sample impact wrenches selected in accordance with 4.3.2 shall be tested as follows to determine compliance with the requirements of this specification. The AQL shall be 2.5 percent defective. Nonconforming sample wrenches shall not be included in the lot offered for delivery and if the number of defective units exceed the acceptable number for the samples tested, the lot represented shall be rejected.

4.5 Tests.

4.5.1 Electromagnetic interference suppression test report. When electromagnetic interference suppression is required (see 3.8.4.4), the manufacturer shall furnish: (1) a certified test report showing that the wrenches supplied have been tested and are within the limits specified for portable electric tools in accordance with the requirements of MIL-STD-1337 or (2) certification that the electromagnetic interference suppression network in the wrenches offered is identical to that of wrenches previously tested and accepted. The electromagnetic interference suppression system shall be disconnected during the tests of 4.5.2.

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4.5.2 Motor and electrical equipment tests. Each sample impact wrench shall be tested to determine compliance with the requirements of UL45 for industrial tools, for the following:

(a) Continuity of grounding connections.
(b) Dielectric strength.
(c) Current input (no-load).

4.5.3 No-load test. Each impact wrench shall be operated at no-load for a period of one hour (thirty minutes in the forward, and thirty minutes in the reverse directions) during which time it shall be checked for switch operation, excessive noise, virbration, indication of overheating and low or excessively high spindle speed (RPM). No-load r.p.m. shall not vary more than ± 10 percent.

4.5.4 Performance. The impact wrench shall be subjected to a performance test to determine conformance with the torque, free speed, and impact rate requirements of table I. The torque test shall be conducted on an analyzer of suitable capacity and accurate calibration to insure correct results.

4.5.5 Wrench components test. The impact wrench shall satisfactorily withstand a test consisting of operation for 4 hours in cycles of 10 seconds "ON" and 10 seconds "OFF" under the load conditions specified in UL45 for impact wrenches. The direction of spindle rotation shall be reversed after each 10 minutes cycling. At the completion of the test run, the wrench shall be disassembled for inspection, and shall show no damage, lubricant leakage nor evidence of failure. In addition to the components test, current input and motor temperature rise shall also be determined.

4.6 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage shall be inspected to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40 for Wrench, Impact, Electric.

6. NOTES

6.1 Intended use. Impact wrenches furnished under this specification are intended for electric power driving of sockets, screwdriver bits and installing and removing nuts, bolts and screws.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2 and 3.7).
(c) Special spindle end and size when required (see 3.8.3).
(d) Cord (cable) length, if different (see 3.8.4.3).
(e) Electromagnetic interference suppression when required (see 3.8.4.4. and 3.3.1).
(f) Fungus protection when specified (see 3.8.4.5).
(g) Carrying case, if required (see 3.10).
(h) Handbook of instruction or parts list, if required (see 3.12).
(i) Kind and quantity of accessories required (see 3.13).
(j) Spare parts when required (see 3.14).
(k) First article inspection when specified (see 3.2).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 First article approval. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article approval as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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MILITARY INTEREST:

Custodians:

Army - GL
Navy - SH
Air Force - 84

Review activities:

Army - GL
Navy - SH
Air Force - 84

User activities:

Army - AV, EL
Navy - AS, OS, MC, YD

Preparing activity:

Navy - SH

CIVIL AGENCY INTEREST:

GSA-FSS

Project No.(5130-0062)

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION W-W-650 C - WRENCH, IMPACT, ELECTRIC

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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